

# PAUL MITCHELL

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## ARTIFICIAL INTELLIGENCE ENGINEER · AGENTIC SYSTEMS & AI SAFETY

### PROFESSIONAL SUMMARY

William & Mary M.S. in Computer Science candidate (3.95 GPA, AI/ML concentration) and clearable U.S. citizen targeting Booz Allen Hamilton's AI Engineering & Agentic Systems practice. Bridges applied machine learning, multi-agent RAG architectures, and adversarial AI safety with hands-on systems engineering, enterprise FastAPI microservices, and 100+ node Linux infrastructure telemetry. Committed to building robust, secure, and explainable AI solutions for critical mission missions.

### EDUCATION

College of William & Mary — M.S. in Computer Science (AI/ML & Systems)

Expected Dec 2026 · GPA: 3.95

College of William & Mary — B.S. in Computer Science

Graduated Dec 2024

Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

### TECHNICAL SKILLS & INFRASTRUCTURE

|                              |                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------|
| Agentic AI & LLM Systems:    | LangGraph, LangChain, RAG architectures, vector databases (FAISS, Milvus), prompt engineering, LLM guardrails |
| Machine Learning & NLP:      | PyTorch, Hugging Face, DistilBERT, Sentence-BERT, BERTopic, LightGBM, scikit-learn, model evaluation          |
| Backend & Cloud Engineering: | Python, FastAPI, REST APIs, Docker, AWS (S3, EC2, Bedrock), Linux (Ubuntu/RHEL), Git/GitLab CI, SQL           |
| Systems & Observability:     | Prometheus, Grafana, telemetry instrumentation, NIST AI Risk Management Framework, automated evaluation       |

### TECHNICAL & PROFESSIONAL EXPERIENCE

Data Analytics & AI Extern | Extern, Amazon-Sponsored Project | Remote

Jun 2025 – Sep 2025

- Engineered end-to-end NLP pipelines combining fine-tuned DistilBERT ( $F1 = 0.89$ ) and BERTopic clustering on ~20,000 unstructured text records, automating feature extraction and topic modeling.
- Built a predictive classification engine using LightGBM and deployed an interactive Streamlit application delivering real-time decision-support analytics and metric visualization for leadership.
- Prototyped an LLM retrieval insight interface querying structured text corpora with grounded citations, enforcing deterministic safety guardrails.
- Implemented data validation pipelines on AWS S3 storage ensuring consistent feature representation and provenance across multi-team evaluations.

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA

Feb 2025 – Present

- Administer 100+ Linux instructional compute servers, configuring networking, DNS, and user isolation via Puppet.
- Deployed Prometheus and Grafana telemetry dashboards monitoring compute cluster utilization, hardware memory pressure, and node health, accelerating incident triage and infrastructure reliability.
- Evaluated multi-tenant container isolation trade-offs, documenting operational and security guidelines for campus compute clusters.
- Engineered automated system verification scripts that diagnosed and resolved recurring network route bottlenecks.
- Maintained comprehensive hardware lifecycle inventories and license mappings, ensuring compliance across 100+ high-availability nodes.

### KEY TECHNICAL & SYSTEMS PROJECTS

ShadowPrompt: Adversarial LLM Red-Teaming & Honey-Prompt Defense | Python, FastAPI

2026 · [pjmittchell7.github.io/shadowprompt](https://github.com/pjmittchell7/shadowprompt)

- Engineered an adversarial fuzzer simulating steganographic token smuggling (zero-width payloads and prompt injection) against defense-grade LangGraph multi-agent RAG pipelines.
- Built an air-gapped pre-inference security proxy catching 99.2% of attacks in < 1ms; engaged honey-prompt sandboxes to neutralize data exfiltration and enforced citation grounding under NIST AI Risk standards.

PromptShield: Tokenization-Aware Adversarial LLM Firewall | Python, FastAPI, scikit-learn

2026

- Engineered a pre-inference security proxy intercepting prompt injection and jailbreak attacks prior to model execution; benchmarked at 0.27-1.11 ms scoring latency with zero model retraining required.
- Maintained false positive rates under 0.1% across complex adversarial prompts, code evaluation benchmarks, and multi-turn dialogues.

Healthcare Network Traffic Anomaly Detection Framework | Python, scikit-learn

2025

- Developed an automated network packet inspection pipeline featuring feature extraction and threshold calibration to support SOC operational alerting.
- Benchmarked unsupervised anomaly detection algorithms against high-volume network telemetry to eliminate false positive alert fatigue.